

CO544 - Machine Learning Lab 2 - Report

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PRELIMINARIES

1. Objective

To study and verify the basic properties of vectors and matrices using Python, and to analyze the specific behavior of symmetric matrices and their eigenvectors.

2. Summary of Results and Descriptions

Vector Operations

- **Dot Product and Angle:** The dot product is 0 and the angle is 1.5707 radians (90). This shows that vectors x and y are **orthogonal** (perpendicular to each other).
- **Vector Norm:** The total length (norm) of vector x is 2.236.

Matrix B Properties

- **Symmetry Check (B - B.T):** The result is **all zeros**. This proves that Matrix B is a symmetric matrix ($B = B^T$).
- **Trace and Determinant:** The trace is 18.0 (the sum of the middle diagonal numbers) and the determinant is 65.0.
- **Matrix Multiplication:** Multiplying Matrix B with a random vector z gives a new vector with a shape of (3,). The quadratic form calculation gives a single number output of 11.721.

Eigenvalues and Eigenvectors

- **Eigenvalues:** Matrix B has three different hidden values (eigenvalues): 13.117, 3.443, and 1.439.
- **Eigenvector Dot Product:** The dot product of the first two eigenvectors is -2.775e-16. This scientific number is practically **0** (the tiny difference is just a computer rounding mistake). This proves that the eigenvectors are also perpendicular to each other.

Question: Can you formally prove that the dot product of the eigenvectors is expected to be zero for the specific structure in matrix B?

Proof:- Matrix B is real symmetric $B = B^T$. Let u_1 & u_2 are 2 distinct eigenvectors of B. Eigenvalues are λ_1 and λ_2 .

$$Bu_1 = \lambda_1 u_1 \quad \text{--- (1)}$$
$$Bu_2 = \lambda_2 u_2 \quad \text{--- (2)}$$
$$(Bu_1)^T = (\lambda_1 u_1)^T$$
$$u_1^T B^T = \lambda_1 u_1^T$$

Since B symmetric, $B^T = B$

$$u_1^T B = \lambda_1 u_1^T$$

Then, use equation by u_2

$$(u_1^T B) u_2 = \lambda_1 u_1^T u_2 \quad \text{--- (3)}$$
$$u_1^T (Bu_2) = \lambda_2 u_1^T u_2 \quad \text{--- (4)}$$

Because matrix multiplication associative

$$\lambda_1 u_1^T u_2 = \lambda_2 u_1^T u_2$$
$$\lambda_1 u_1^T u_2 - \lambda_2 u_1^T u_2 = 0$$
$$(\lambda_1 - \lambda_2) (u_1^T u_2) = 0$$
$$(\lambda_1 \neq \lambda_2) \Rightarrow \therefore \lambda_1 - \lambda_2 \text{ cannot be zero}$$
$$u_1^T u_2 = 0$$

$\therefore$ This proves that any two distinct eigenvectors of a symmetric matrix are always orthogonal as well.

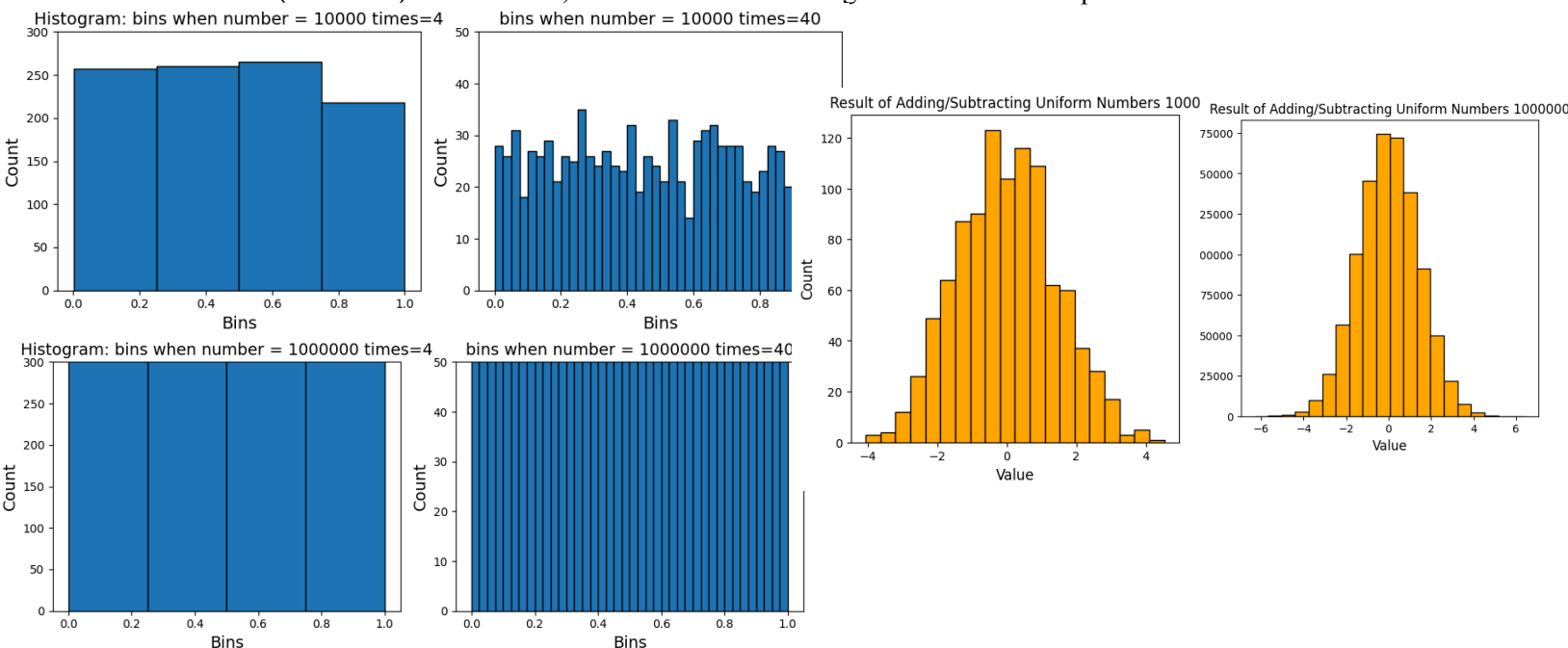
1. Random Numbers and Uni-variate Densities

1. Uniform Distribution Questions - Blue

- **Why the histogram is not perfectly flat:** Because 1,000 numbers is a small sample. Random chance causes small statistical variations, making some bars slightly taller than others.
- **Why it looks different every time you run it:** Because the computer generates a completely new set of independent random numbers during each run.
- **How observations change with more data (ex- 100,000 numbers):** The random variations balance out, and the histogram bars become perfectly flat and equal.

2. Adding and Subtracting Questions - Yellow

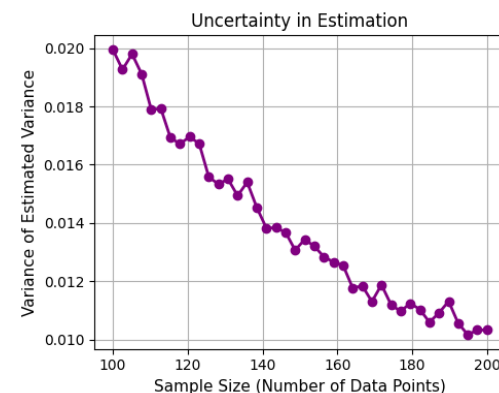
- **What do you observe?** The histogram forms a symmetrical, bell-shaped curve centered around 0. It is no longer flat.
- **How it changes when you change the amount of numbers you add/subtract:** If you add/subtract more numbers, the curve becomes even smoother and closer to a perfect bell curve. If you use fewer numbers, the shape looks rough and less clear.
- **Is there a theory that explains this?** Yes. The **Central Limit Theorem (CLT)**. It states that when you add many independent random numbers together, their sum will always form a **Normal (Gaussian) Distribution**, no matter what their original distribution shape was.



2. Uncertainty in Estimation

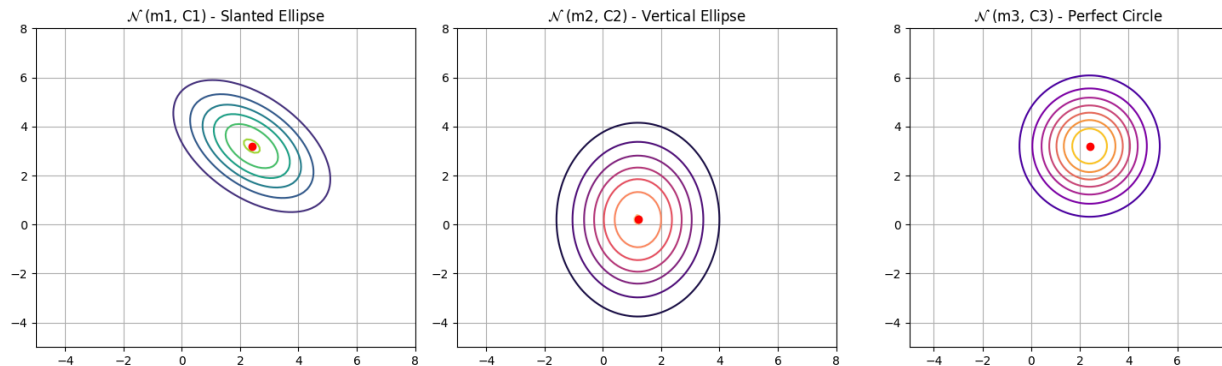
Discussion

- **What the Plot Shows:-** The vertical axis measures how wildly our variance estimates change from trial to trial. The downward curve proves that as the Sample Size increases, this trial-to-trial bouncing decreases drastically.



- When data points are limited, a few random outliers can heavily skew your parameter estimation. Gathering more data allows random errors to naturally cancel out.
- Machine Learning Context:- This experiment visually confirms a fundamental rule of machine learning: Larger datasets directly decrease parameter uncertainty, making model training far more stable, precise, and reliable.

3. Bi-variate Gaussian Distribution



Discussion

- **Role of the Mean:** The mean tells us where the center of the distribution is. It moves the highest peak of the 3D mountain to a specific point on the map. Changing the mean only moves the mountain to a new location, it does not change its shape or width.
- **Role of the Diagonal Numbers (Variance):** These numbers control how wide or stretched the shape is along the grid lines. For example, when the Y-axis number is bigger than the X-axis number, the shape stretches out tall and long along the Y-axis. When both numbers are exactly the same and independent, the shape spreads out equally in all directions, making a perfect circle at the bottom.
- **Role of the Corner Numbers (Covariance):** -These numbers show if the two directions are connected to each other. If these numbers are zero, it means the two directions are independent, so the shapes stand straight up and down along the grid lines. If we put a non-zero number (like -1), it tilts the whole distribution, making the 3D mountain slant diagonally.

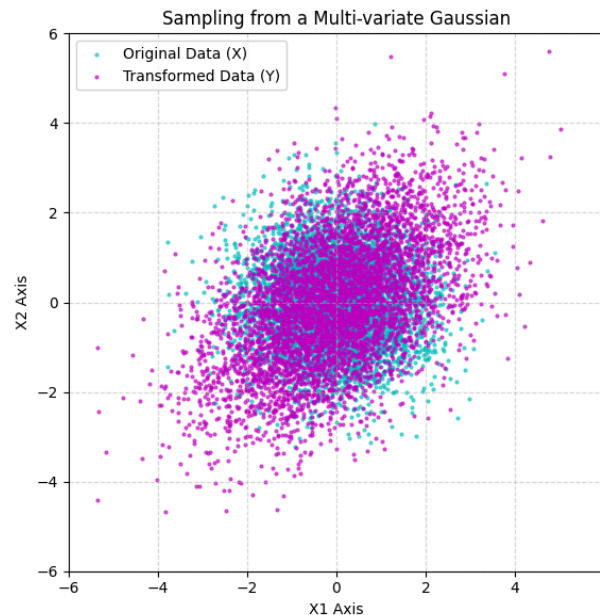
4. Sampling from a multi-variate Gaussian

Observations

- **Original Data (X - Cyan):** The original data points are distributed in a perfect **circular shape** centered around (0,0). This shows that the initial data points are independent and have equal variance along both axes.
- **Transformed Data (Y - Magenta):** After multiplying the original data by the matrix A, the shape changes into a **slanted ellipse**. The points are now stretched diagonally from the bottom-left to the top-right.

Discussion

- **Why the shape changed:** The original data X has no correlation. When we multiply it by matrix A (which comes from the covariance matrix C), it injects correlation into the data.
- **The Result:** This matrix multiplication stretches and rotates the circular data distribution to match the exact shape and orientation defined by our target covariance matrix.



5. Distribution of Projections

Results Summary

- **Maximum Value on Graph (Maxima):** Around **3.00**.
- **Minimum Value on Graph (Minima):** Around **1.00**.

Short Discussion

- **What the Graph Shows:** The red wave graph shows what happens to the variance of the data when we look at it from different angles (θ). Because the data cloud is shaped like a slanted egg, the variance naturally changes smoothly up and down like a wave as we turn our viewing angle.
- **The Relationship with Eigenvalues:** The largest eigenvalue of our covariance matrix is exactly **3.00**, and the smallest eigenvalue is exactly **1.00**. This perfectly matches our graph:
 - ◆ The highest peak of the wave (**maxima**) reaches exactly **3.00** (the largest eigenvalue).
 - ◆ The lowest point of the wave (**minima**) drops exactly to **1.00** (the smallest eigenvalue).
- **The Relationship with Eigenvectors:** The eigenvectors show the exact directions where these maximum and minimum points happen. The first eigenvector points along the longest direction of the data (giving the max variance of 3.00), and the second eigenvector points along the shortest direction of the data (giving the min variance of 1.00).

3. Analytical Confirmation (Why it looks like a Sine Wave)

We can prove mathematically why the graph looks like a smooth sine wave. When we project our data onto a direction vector u , the formula for the variance is:

$$\text{Variance} = u^T C u$$

Using our direction vector $u = [\sin \theta, \cos \theta]^T$ and our covariance matrix $C = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, multiplying the matrices gives:

$$\text{Variance} = 2 \sin^2 \theta + 2 \cos^2 \theta + 2 \sin \theta \cos \theta$$

Using simple trigonometry rules,

$$\text{Variance} = 2 + \sin(2\theta)$$

This mathematical formula analytically confirms why the graph is a perfect **sinusoidal wave** that swings between a maximum of $2 + 1 = 3$ and a minimum of $2 - 1 = 1$!

